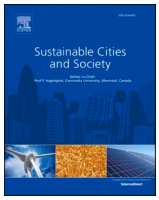
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A novel hybrid systemic modeling into sustainable dynamic urban water metabolism management: Case study

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ABSTRACT

Over the past few years, due to increasing population, limited water resources, and other pressures on the urban water systems (UWS), urban water management has become a major concern for urban policymakers to develop efficient management solutions. In this study, a new hybrid model called sustainable dynamic urban water metabolism management (SDUWMM) is presented, based on urban water metabolism and system dynamics. Isfahan's UWS has been modeled by SDUWMM and its conditions have been simulated until 2041. SDUWMM considers effective internal relationships of the UWS components such as population, climate change, water supply and demand, infrastructures condition, water price, costs and revenues and social behaviour. The model analyzed impacts of increasing water price, capacity building and social awareness, drought, change in rehabilitation rate and installing conserving fixtures on debt and water shortage. According to the study results, due to high inflation rates and annual rise in costs, achieving financial sustainability is quite challenging. Under optimum conditions, the debt can be reduced up-to 30%, but water shortage can be reduced up-to 200% in 2041. SDUWMM helps a better understanding and more holistic assessments of the performance of UWS for Isfahan's water supply decision-makers, and its methodology is applicable to other cities.

1. Introduction

Cities and societies worldwide face the water management as a main challenge for urban management. According to United Nations Sustainable Development Goal (UN SDG), urban water management is an important factor for the formation of a sustainable society (Sørup et al., 2020; Unaited Nations, 2015). The sustainable development of urban environments demands water resources, that are vital for supporting societies and socio-economic activities. Mismanagement of water will result in adverse effects on human health, economy, and environment. In addition, approximately 70% of the world population will live in urban areas by 2050 (United Nations, 2018) . However, most cities are currently facing with the water security challenge. They are under water stress and do not have sustainable water resources to meet the human society's needs. It is also expected that the climate change will further complicate the issue of meeting the needs of future generations (Ramôa et al., 2015). This condition require sustainable solutions for both water demand and supply. Thus, having access to reliable and sufficient water sources to sustain the demand in quality and quantity has a critical role for human life and societies (Feizizadeh et al., 2021; Yang et al., 2017).

In recent years, an improvement in the society living standards has led to a dramatic increase in commodities consumption and unnecessary services. Caused by the larger world population, the increasing use of resources by cities has created a pressure on natural resources. These changes have also affected the pattern of water demand and increased urban water consumption, especially for sanitary purposes (Ramôa et al., 2015; UNESCO, 2009).

Various factors cause pressure on UWS, which can be divided into three categories: (a) Social Pressures (population growth, urbanization, Change in Lifestyle, Preferences, and Confidence in Services), (b) Environmental Pressures (Water Stresses (Quantity & Quality), climate change) and (c) Economical Pressures (Escalating and Variable Costs, Tariff Structures for Water Services, Competition and Political Conflicts, Financing Difficulties) (Howe et al., 2011; Marlow et al., 2013; Novotny et al., 2010; Polonenko et al., 2020; Ramôa et al., 2015; Segrave et al., 2007; UN-DESA, 2009; UNESCO, 2009).

According to these pressures, conventional UWS, which are mostly designed by engineers and are not compatible with newer water

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management challenges, cannot play their effective roles in the future. The main focus of the conventional urban water management paradigm is the development of centralized infrastructure that is practically implausible and prevents the implementation of new approaches to sustainable urban water management (Brown et al., 2009). Due to progressive water problems as a result of social, environmental, and economic pressures, decision-makers need to modify urban water management strategies to address upcoming issues, including water stress, climate change, population growth, and movement toward a sustainable approach (Ding & Ghosh, 2017; Hashempour Bakhtiari et al., 2020). Achieving a sustainable development approach requires a shift in the attitude of decision-makers. They should adopt the most appropriate strategy by considering different conditions. According to the goal of urban water management, sustainability of this system is defined in the form of safe, sufficient and cost-effective water supply for all clients without affecting the lives of future generations under an efficient government and efficient infrastructure (Howe et al., 2011; Novotny et al., 2010).

To achieve sustainable urban water services, it is essential to have a long-term vision of the service and consider all dimensions of sustainability (van der Zouwen et al., 2015).

There are various methods for management of urban water system, such as integrated urban water management, multi-objective decision-making, System Dynamics (SD), Life Cycle Assessment (LCA), and Urban Water Metabolism (UMW), each of which have their own features and they are applicable for specific purposes. For example, LCA determines the amount of resources used in goods and services (such as energy and water footprints) in an urban system which considers the past (Bach et al., 2014). The SD approach describes relationships of complex systems over time through internal feedback cycles, delays, and nonlinear connections and can simulate the dynamic behaviour of system (Correia de Araujo et al., 2019; Sterman, 2000). The integrated water cycle modeling considers a water system in an urban unit such as the district, without considering the entire urban area or the entire economy (Bach et al., 2014). According to the goals of this article, which is the development of a framework for simulation of urban water system management measures to achieve the system's sustainability. In this regard, a novel framework consist of two methods of UWM and SD has been used. The following are the features of these two methods and the benefits of their hybridization.

Urban metabolism approach has developed to identify critical processes that may threaten sustainable urban management. This approach helps urban policymakers to improve the usage of environmental resources and capacities to deliver goods and services. The urban metabolism approach provides an extensive image of input and output flows (water, waste, nutrients, energy, etc.). Also, their quantitative analysis that can be used as a conceptual and analytical framework (Farooqui et al., 2016; Kennedy et al., 2011; Renouf et al., 2016; Wolman, 1965). Despite the growth of research on the topic of urban metabolism sources, relatively little attention has been paid to water (Kennedy et al., 2007). Some of the previous studies that investigated urban water flow have provided incomplete images of water metabolism. Binder et al. (1997) developed a water metabolism model for a city in Colombia, but faced many problems with the availability and poor quality of data. Kenway et al. (2011) presented a comprehensive report on UWM. They surveyed several cities in Australia and found that large streams of rainwater, stormwater, and sewage have remained unused. Although, the developed UWM framework in their study was based on water mass balance and has been employed in various studies in recent years (Ghosh et al., 2019; Paul et al., 2018; Singkran, 2017), the water losses were ignored in the proposed model, which is an important component of urban water balance in a developing city.

The strength of UWM approach relies on a comprehensive view and accuracy in hydrological mass balance (Renouf & Kenway, 2017). The UWM approach can provide a comprehensive framework of urban water management by measuring all water flows in the system and create a big picture of urban water management to evaluate water efficiency in the urban water system (Renouf et al., 2017). Determination of urban water management performance is important to monitor and assess the progress of urban water management goals. Renouf and Kenway (2017) pointed to the absence of a comprehensive approach to evaluating the water management performance at macro urban scale, and stated that the UWM approach was the most useful to in achieving this.

Urban water models, such as Aquacycle and Urban Water Optioneering Tool (UWOT), work based on water balance and focus on the hydrology of flows that are managed by urban infrastructure on a city-scale (Jeong & Park, 2020) and do not consider social and economic dimensions of the city.

On the other hand, the creation and calculation of water metabolism is inherently complex. As water is flowing through a dynamic hydrological cycle, natural water flow data, especially surface runoff, evapotranspiration, and groundwater infiltration, must enter the metabolism model using other studies and modeling (Paul et al., 2018). Also, UWM is a static approach, so the dynamic feature of the natural water flow in the model is simplified and ignored. To overcome this limitation, the UWM approach can be combined with other approaches. Simulation results from UWS modeling can be incorporated in urban metabolism. In addition, water flow data from UWM can be used for System Dynamics (SD) analysis.

The UWS is inherently a dynamic system consisting of several interconnected subsystems that are influenced by each other. Therefore, it is helpful to design a systematic and dynamic approach to its management (van der Zouwen et al., 2015). SD models have empowered decision-makers to implement the most optimal water management strategies via assessing the dynamic behavior of UWS components (Pluchinotta et al., 2021; Xu et al., 2020; Zarghami & Akbariyeh, 2012).

Rehan et al. (2013) developed causal loop diagrams and an SD model for financially sustainable management of urban water distribution networks. This model was comprised of water network, consumer, and finance sectors. In subsequent studies, Sahin et al. (2015) and Sahin et al. (2016) developed urban water management models that applied the SD approach to create price adjustments, which could, in turn, generate revenue to invest in desalination plants capable of providing water security efficiently in the future. Correia de Araujo et al. (2019) built an SD model to support water management and focused on demand management and water scarcity in semiarid regions. This model took account of population, water supply, water demand, water tariff, and return water sub-models. Xu et al. (2020) established a bi-objective model to balance economic and environmental costs of urban water supply system. This study considered household and industrial water demand and applied an SD model to test water supply system performance under short, medium, and long-term planning scenarios. This model did not consider some of the essential components and relations of UWS.

Zhang et al. (2021) proposed an assessment for energy-water metabolic network in province scale in China. In this model, input-Output analysis and ecological network analysis were integrated through the SD to clarify embodied linkages in energy-water nexus. This model has provided a new framework for evaluation, yet with limited time span, has not provided predictions and suggestions on future actions, and has ignored a large part of economic and social parameters. Pluchinotta et al. (2021) investigated sustainable urban water management in Ebbs fleet by developing a participatory SD model. The purpose of this study was to investigate the methods of reducing potable water consumption through interventions such as socio-environmental and economic incentives. Also, they suggested an improvement in the performance of SD models, and integrated it with quantitative tools like UWM modeling to provide more detailed results of impacts of different strategies on urban water management.

This paper presents a new hybrid urban water management model (SDUWMM), based on the integration of SD and UWM in Isfahan, Iran, to analyze and simulate the UWS interactions dynamically and

comprehensively. Combining SD with UWM improves its quantity and accuracy and the use of SD along with UWM improves the dynamic features' analyses of the system and better display changes.

The effect of managerial measures and interactions between components in UWM are displayed with more dynamic details. SDUWMM includes various aspects of UWS such as water supply and demand, infrastructure conditions, water prices, cost and revenue of water companies, and social behavior. The impacts on cost coverage and water shortage from eight scenarios in four major aspects were examined: Economics (increasing water price), Social (Capacity building and social awareness), Environmental (Climate change and Drought period), and Technical (Change in Rehabilitation rate, Installing conserving fixtures)

The aim of this study was to introduce SDUWMM as a new hybrid model to improve water and financial resources efficiency in urban water management to increase the cost coverage and reduce the gap between water supply and demand. Due to limited financial resources, the new economic flow alongside the water flow in UWM has been defined for the first time, which intends to examine water consumption management and financial resource management methods. In addition, it evaluates economic, technical, environmental and social efficiency of various measures so that decision-makers can benefit from limited financial resources to carry out the most effective measures.

In the following, the Isfahan city case study, was first presented, and then the stages of SDUWMM modeling was described and the model structure outlined. Section 4 explains model simulation and discusses the results. The paper was closed with concluding remarks.

2. Study region

Isfahan city is the third most populated city in Iran after Tehran and Mashhad. It has dry and semi-arid weather. Its average precipitation is 130 mm/year, most of which in winter and early spring (Murray-Rust et al., 2000). Zayandeh-Rud River originates from Zardkouh Bakhtiari Mountains in Zagros Mountain Strings and finds its way into the center of the Iranian plateau, and the city has been created and expanded on both sides. According to the 2016 census in Isfahan, it has 1,961,260 people and a total of 619,091 households (Table 1) (Statistical Centre of Iran, 2016). Fig. 1 shows the position of city of Isfahan in the Zayandeh-Rud River basin in the center of Iran.

In the last few decades, this city has witnessed a rapid population increase due to economic growth and industrial development, tourism, welfare and cultural development. In addition, climate change, decreasing rainfall and increasing costs have caused an imbalance between water supply and demand in this region. In this way, in most recent years, providing urban and industrial water has been accompanied by pressure on agricultural and environmental sectors (Safavi et al., 2015; Etaat & Salehian, 2020) River is the most important river in central Iran and Isfahan region. This river is the main source of water supply in Isfahan. The water in the Zayandeh-Rud Basin is consumed in three major sectors of agriculture, industry and urban, and in addition, environmental needs of the river and Gavkhouni wetland (Ahmadi et al., 2020).

Isfahan water and wastewater company (IWWC) is responsible for water supply and water and wastewater treatment in Isfahan. Approximately 93% of the city's drinking water is provided from Zayandeh-Rud dam and the rest of the water is provided from wells (Isfahan Municipality, 2018). Therefore, focusing on Zayandeh-Rud River is important as the main source of water supply in Isfahan. Surface water resources in this basin include the natural flow of Zayandeh-Rud, the transitional water of the first, second and third Koohrang Tunnels and the Cheshmeh-Langan tunnel (Etaat & Salehian, 2020).

Water allocation is prioritized to urban demand and industry, and then agriculture and environment respectively. The agricultural water allocation of Zayandeh-Rud Basin has fluctuated in recent years, with an overall falling trend, and recently there are no water allocations from Zayandeh-Rud dam for these demands. The continuation of drought conditions in the past years has been so critical in this region that has caused social unrest in fall, 2021, in Isfahan. As a result, attention to the water resources conditions and urban water management should be the main priorities of decision-makers in Isfahan. Table 2 shows the some water management characteristics of study region.

Table 1
Area and population of Isfahan city (Statistical Centre of Iran, 2016).

Item	Area (km^2)	Population (person)
Iran	1,627,771	79,926,270
Isfahan Province	107,019	5,120,850
Isfahan county	15,706	2,243,249
Isfahan City	550.72	1,961,260
% Of Province	0.5%	38.3%
% Of Country	0.03%	2.45%

3. Stages of SDUWMM modeling

In this study, a new hybrid model has been developed from the combination of SD and UWM approach. In the proposed SDUWWM model using a systematic method, a model of influential factors in urban water management has been developed using SD modeling. Utilizing the UWM approach, that is a quantitative framework of model, objectives of the model have been investigated in order to improve the state of sustainable urban water management. In an urban water management model, to establish sustainability in the system, the management of its components should be integrated and systematic. The proposed urban water model examines social, economic, environmental, technical and governmental dimensions, each of which has their own variables and specific characteristics and together form UWM sustainability.

Sustainability in the UWS occurs when the quality of governance and assets were in good condition and actively satisfied water sector's needs securely to meet social, environmental and economic sustainability in the whole city (Venkatesh et al., 2017). In other words, the government regulates roles and relationships according to urban water cycle infrastructure in such a way as to achieve social, economic, environmental, technical, and governance sustainability.

3.1. Urban water metabolism modeling

In this study, the Mass Flow Analysis (MFA) method has been used for UWM analysis. MFA examined mass fluxes of resources (water, material and energy) into a city, the stocks and flows therein, and the mass exchanged with other systems. Moreover, MFA helps decision-makers to understand trends of resource consumption over time (Paul et al., 2018; Renouf et al., 2017). In order to quantify UWM relations, some studies have defined equations (Renouf et al., 2017; Renouf & Kenway, 2017), which are mostly based on the source of water flow. In this study, to develop a more practical, water system compatible operation activities, and easier data collection facility, UWM equations were defined as consumption-oriented. According to the mass balance principle, total input water (resources) should be theoretically equal to total output (consumption) water in the system boundary and UWS processes.

$$AllocableWater = P + SW + \mathrm{GW} + IW + R - E - D \quad \text{(Eq. 1)}$$

$$WaterConsumption = \mathrm{Urb} + Ind + Agr + Env + WL \quad \text{(Eq. 2)}$$

$$AllocableWater \geq WaterConsumption + ChangeinWaterStorage \quad \text{(Eq. 3)}$$

Where:

P: total precipitation falling in the urban boundary

SW: total surface water entered the urban boundary, including river water, dam and seawater

GW: volume of groundwater in the urban boundary

IW: imported water from other regions (treated/raw water)

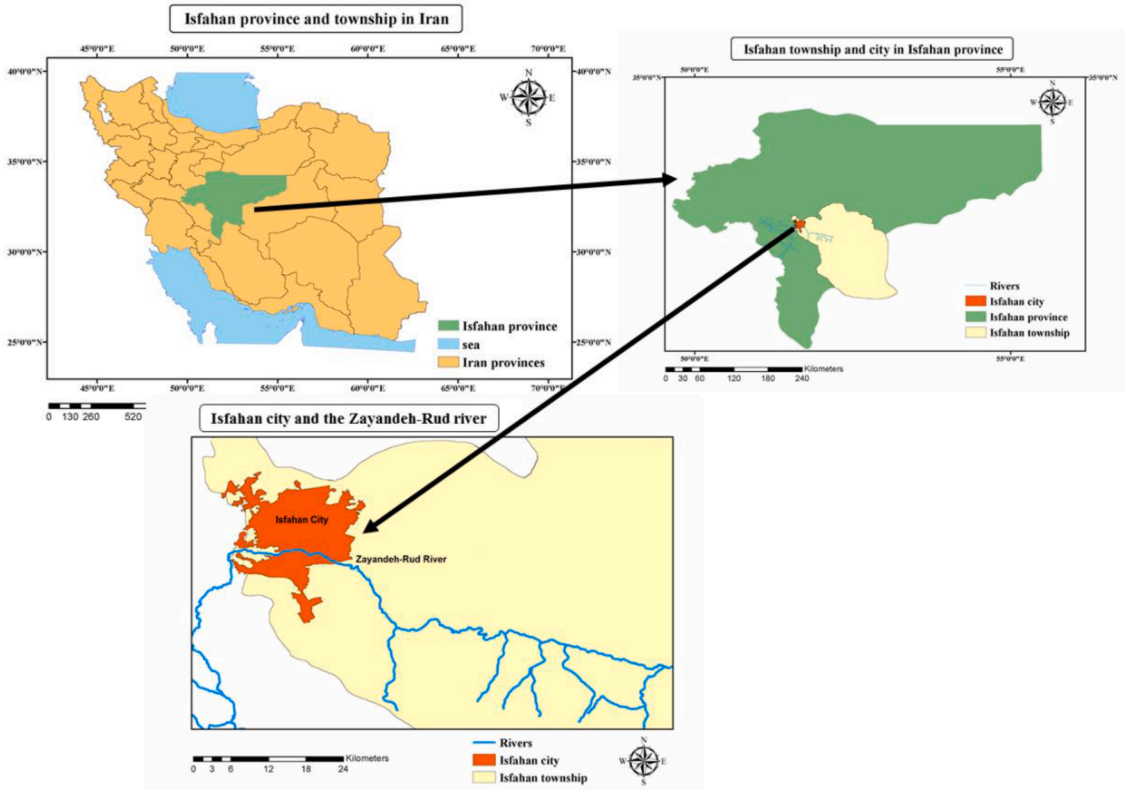


Fig 1. Position of the Isfahan city and Zayandeh-Rud River.

Table 2
Characteristics of study region.

Item	Value
Major water sources	Dam water (Zayandeh-Rud river)
Precipitation	139 mm
Available surface water resources	148 m^3/year/capita
Available groundwater resources	137 MCM/year
Urban water demand	174 MCM
Urban water consumption	146 MCM
Water price	6150 IRR*
Income of Isfahan Water & Wastewater company	90 B IRR
GDP	54.4 B USD

* Iranian Rial

Re: total volume of return water including reuse wastewater and infiltrated water from industrial and agricultural use
E: evapotranspiration from urban areas
D: volume of discharge and outflow water from urban boundary
Urb: urban water consumption
Ind: industrial water consumption
Agr: agricultural water consumption
Env: environmental water demand
WL: water loss from the water system in the urban boundary
In the case of economic flow, the proposed equations are as follows:

$$Fund = TotalIncome - TotalCost \quad \text{(Eq. 4)}$$

$$TotalIncome = \mathrm{Re}v + \mathrm{Bud} \quad \text{(Eq. 5)}$$

$$TotalCost = CC + OC + SC + WP + GC + WC \quad \text{(Eq. 6)}$$

Where:
Rev: revenue from water sales in a year
Bud: government budget for the water sector in one year
CC: capital cost in a year
OC: operation and maintenance cost in a year
SC: staff cost in a year
GC: general cost includes depreciation cost and administrative cost
WC: water conservation cost

3.2. System dynamics modeling

The processes studied in SDUWMM are shown in Fig. 2. This model designed, set up and simulated in the Vensim® software (by Ventana System). It includes climate, water resources, water allocation, population, water demand, water distribution network, cost and revenue of water company and water pricing. One of the features of SD models is the evaluation of theinteractions of processes in the form of feedback which provides more natural behavior close to the reality.

Model accuracy is greatly enhanced by considering UWM. A detailed and secure data bank is essentialfor any integrated urban water management model. Defining relationships in the model, calibration and validation of the model requires accurate, complete and secure data. In

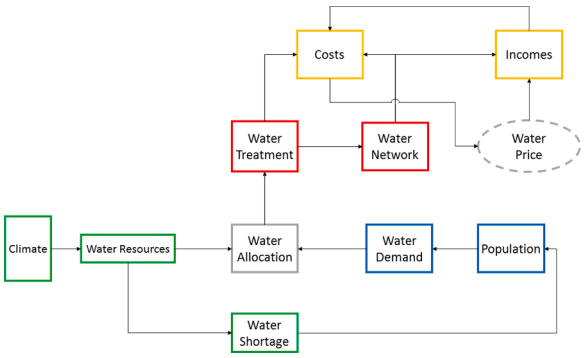


Fig 2. Processes studied in the model.

fact, collecting the necessary data is difficult. In addition, confidentiality and accuracy of information cannot be guaranteed; therefore, data uncertainty is a major challengefacing integrated models. Due to UWS, which includes various subsystems and many internal and external factors, the dynamics of this system will be influenced by the dynamics of smaller components.

The model focuses on a systematic analysis of urban water supply system of Isfahan City. It runs over a time scale of 30 years, accounting for the evolution of the city between 2011 and 2041. The simulation time step is yearly and Input data has been imported to the model annually. This time step is considered due to financial reporting time of IWWC, the annual planning of water system and annual computing of water balance. Different management strategies are defined and implemented on the model to improve its operating status in terms of functional indicators. Since this study is further looking for help with IWWC, it focuses on issues related to urban water under IWWC management. Dynamical relationships of the model were further investigated in greater detail:

3.2.1. Dynamics of available water

An important issue in the Isfahan region is the allocation of water among consumers and setting a balance between water supply and demand. Existing water resources include the natural flow of Zayandeh-Rud River, water transmission from Koohrang 1, 2 and 3 tunnels, and the tunnel of the Cheshmeh-Langan, groundwater resources, returned water from urban, industrial and agricultural consumptions, and precipitation, as shown in Eq. (7).

$$W_{t+1} = W_t + SW_t + GW_t + IW_t + \mathrm{Re}_t + P_t - Urb_t - Ind_t - Agr_t - Env_t - WT_t - E_t - D_t \quad \text{(Eq. 7)}$$

Where:

W_{t+1}, W_t: Water resources available in year t+1 and t, respectively.

SW_t: Input Surface water of Zayandeh-Rud River in year t

GWt: Volume of water provided from water wells in the area in year t

IWt: Imported water flow to the region in year t

Ret: The returned flow of water from urban, industrial and agriculture consumption in year t

Pt: Volume of precipitation in the year t

All parameters are in millions of cubic meters (MCM).

3.2.2. Dynamics of water consumption

Water demand is determined every year based on the population and per capita water consumption of that year. Isfahan's population has experienced a sharp rise in recent years and the rate of increase in water consumption is quite evident over time. The equation of dynamic changes in the annual water demand is shown in Eq. (8).

$$Dw_t = P_t \times q_t \times 365 \quad \text{(Eq. 8)}$$

Where:

Dw_t: Demand water in year t.

P_t: Population of Isfahan city in year t.

q_t: Per capita water consumption per day in year t (m^3 per capita per day).

Some of the proposed management strategies in this study focus on reducing water consumption per capita in order to control water demand. In the following sections, the results are presented.

3.2.3. Dynamics of cost coverage index

The cost coverage index (CCI) indicates the amount of cost compensation through revenues. Greater than or equal to one indicates the desirable status of the IWWC's revenue and system economical sustainability. As shown in Eq. (9), this indicator is obtained from the division of annual revenues at the annual cost of IWWC.

$$CCI_t = \frac{R_t}{C_t} = \frac{\mathrm{Pr}_t \times Rw_t + Bud_t}{CC_t + OC_t + SC_t + WP_t + GC_t + WC_t} \quad \text{(Eq. 9)}$$

Where:

Rt: revenue of Isfahan water and wastewater company in year t

Prt: average price of water and subscription in year t

Rwt: volume of revenue water in year t

Bud_t: government budget of IWWC for water section in year t

Ct: total cost of IWWC in the water sector in year t

CCt: capital cost of IWWC in the water sector in year t

OCt: operation and maintenance cost of IWWC in the water sector in year t

SCt: staff cost of IWWC in the water sector in year t

WPt: water production cost in year t

GCt: general cost includes depreciation cost and administrative cost of IWWC in year t

WCt: water conservation cost of IWWC in year t

As in Eq. (9), income is the product of water sales and increases with higher consumption and sales of water.

Nevertheless, one of the urban water management policies, due to the limitations of water resources, is a reduction in water consumption, which conflicts with the economic benefits of water supply companies, which should be addressed.

3.2.4. Dynamics of water supply reliability index

The water supply reliability index (WSRI) represents the amount of supplied water from water demand or, i.e., it expresses water scarcity. This index is calculated based on Eq. (10).

$$WSRI_t = \frac{Rw_t}{Dw_t} = \frac{Urb_t - WL_t}{Dw_t} \quad \text{(Eq. 10)}$$

Where:

Rwt: volume of revenue water in year t

Dwt: Demand water in year t

Urb_t: volume of water allocated to urban usage in year t

WLt: water loss in year t

3.2.5. Dynamics of water network infrastructure condition

The water network condition and the age of pipes change over time. Eqs. (11) and (12) show examples of this dynamic behavior for the two groups of pipes with ages 0-14 and 15-24 years.

$$P_{(0-14)_{t+1}} = P_{(0-14)_t} + EP_t + RP_{(25-34)_t} + RP_{(35-45)_t} + RP_{(above45)_t} - \frac{P_{(0-14)_t}}{15} \quad \text{(Eq. 11)}$$

$$P_{(15-24)_{t+1}} = P_{(15-24)_t} + \frac{P_{(0-14)_t}}{15} - \frac{P_{(15-24)_t}}{10} \quad \text{(Eq. 12)}$$

Where:

P(0-14)t, P(0-14)t+1: the length of pipes with ages between 0 and 14 years in year t and t+1, respectively.

EPt: The length of new extended pipes in year t

RP(15-24): The length of pipes with ages between 15 and 24 which have been rehabilitated in year t.

RP(25-34): The length of pipes with ages between 25 and 34 which have been rehabilitated in year t.

RP(35-45): The length of pipes with ages between 35 and 45 which have been rehabilitated in year t.

RP(above 45): The length of pipes with ages above 45 years which have been rehabilitated in year t.

As shown in Eq. (11), the length of pipes of 0-14 years in year t+1 is equal to the length of the pipes of 0-14 years in year t, plus the length of the pipes caused by expansion of the network in year t, plus the length of rehabilitated pipes in year t, minus the length of pipes that run out of 0-14 and enter to 15-24.

3.2.6. Dynamics of greenhouse gas emissions

Determination of the greenhouse gas (GHG) emissions amount in urban water services is difficult because the sources of emissions are complicated. In water supply services, most of GHG emissions come from electricity consumption. In this study, the amount of electricity consumption in water treatment and distribution processes is considered. The equation of dynamic GHG emissions is shown in Eq. (13).

$$GHG_t = Urb_t \times En_t \times 0.3855 \quad \text{(Eq. 13)}$$

Where:

GHGt: GHG emission as Kg CO_2-eq in year t

Urb_t: volume of water allocated to urban usage in year t

Ent: Energy consumed for one cubic meter water in water sector in year t

Table 3 shows information about the variables of the model.

4. Model performance

4.1. Model performance evaluation

SDUWMM model has been validated and revised with structural tests, structure-oriented behavior tests and behavior pattern tests (Barlas, 1996). The model was calibrated after validation using historical data from 2011 to 2019. The performance of the model was investigated using MAPE (mean absolute percentage errors) for various parameters such as water consumption, water price, water production cost, operation and maintenance cost of the water treatment plant, staff cost and water loss. The results showed that SDUWMM simulated all of them with less than 5% error, which is acceptable. Fig. 3 shows the value of observational and simulated parameters.

4.2. Model simulation

Due to increasing water demand in Isfahan and reducing surface water and groundwater resources in recent years, attention is important for improving the state of urban water management in Isfahan and requires corrective actions. The proposed model of Isfahan water metabolism is shown in Fig. 4 (based on 2018 data). If more detailed data were available, the model could be developed in seasonal or monthly periods or separated by urban districts or type of subscribers. This could improve understanding of the decision-makers in the analysis of the water system in the seasons of the year with various water consumption patterns. It is observed that nearly 30 MCM of water is lost in the production and distribution processes. Fig. 5 clearly shows that with the approaching water to the consumption stage, its actual cost increased. Therefore, it seems that water management solutions are more profitable by focusing on demand management. In other words, cost-saving of 1 m^3 water consumption reduction in the end user is equivalent to reducing the cost of providing 30 m^3 raw water. In fact, the volume of water saved is also important in determining the most suitable strategy.

In the following, eight scenarios in the economic, social, technical and environmental sectors are defined and simulated by the SDUWMM model. The results of these scenarios can provide the decision-makers with a better understanding of the dynamic behaviour of the system and help them in choosing strategies for better water management in Isfahan. In this study, the conditions in Isfahan under different scenarios for the years 2021 to 2041 were investigated.

4.2.1. Scenarios

4.2.1.1. Base scenario. In this scenario, water prices will increase by 7% annually, and the water price elasticity is zero. Social awareness and installing water conservation equipment are not considered. The rehabilitation rate of water network is 0.75% annually (BAU). The modeling results are shown in Figs. 6 and 7. As seen in these graphs, in the coming

Table 3
Variables of the SD model.

Variable	Abbrev.	Units	Source(s)	Variable source type
Water resources available	W	MCM	Iran Ministry of Energy	Modeled
Surface water	SW	MCM	Iran Ministry of Energy	Statistical
Groundwater	GW	MCM	Iran Ministry of Energy	Statistical
Imported water	IW	MCM	Iran Ministry of Energy	Statistical
Returned flow	Re	MCM	Iran Ministry of Energy	Modeled
Precipitation	P	MCM	Isfahan meteorological organization	Statistical
Demand water	Dw	m^3	Isfahan statistical yearbook	Modeled
Population	P	person	Statistical Center of Iran	Modeled
Per capita water consumption	q	m^3/person/day	Isfahan Water & Wastewater Company	Modeled
Cost coverage index	CCI	Dmnl	Survey data	Modeled
Revenue of IWWC	R	IRR*	Isfahan Water & Wastewater Company	Statistical
Average price of water and subscription	Pr	IRR/m^3	Isfahan statistical yearbook	Modeled
Revenue Water	Rw	m^3	Isfahan Water & Wastewater Company	Statistical
Government budget	Bud	IRR	Isfahan Water & Wastewater Company	Modeled
Total cost	C	IRR	Isfahan Water & Wastewater Company	Modeled
Capital cost	CC	IRR	Isfahan Water & Wastewater Company	Modeled
Operation cost	OC	IRR	Isfahan Water & Wastewater Company	Modeled
Staff cost	SC	IRR	Isfahan Water & Wastewater Company	Modeled
Water production cost	WP	IRR	Isfahan Water & Wastewater Company	Modeled
General cost	GC	IRR	Isfahan Water & Wastewater Company	Modeled
Water conservation cost	WC	IRR	Survey data	Modeled
Water supply reliability index	WSRI	Dmnl	Survey data	Modeled
Water allocated to urban usage	Urb	m^3	Isfahan Water & Wastewater Company	Modeled
Water Loss	WL	m^3	Isfahan Water & Wastewater Company	Modeled
Length of new extended pipes	EP	Km	Survey data	Modeled
Greenhouse gas emissions	GHG	Kg/year	Survey data	Modeled
Energy consumption	En	Kwh/m^3	(Wakeel et al., 2016; Zhang et al., 2021)	Modeled

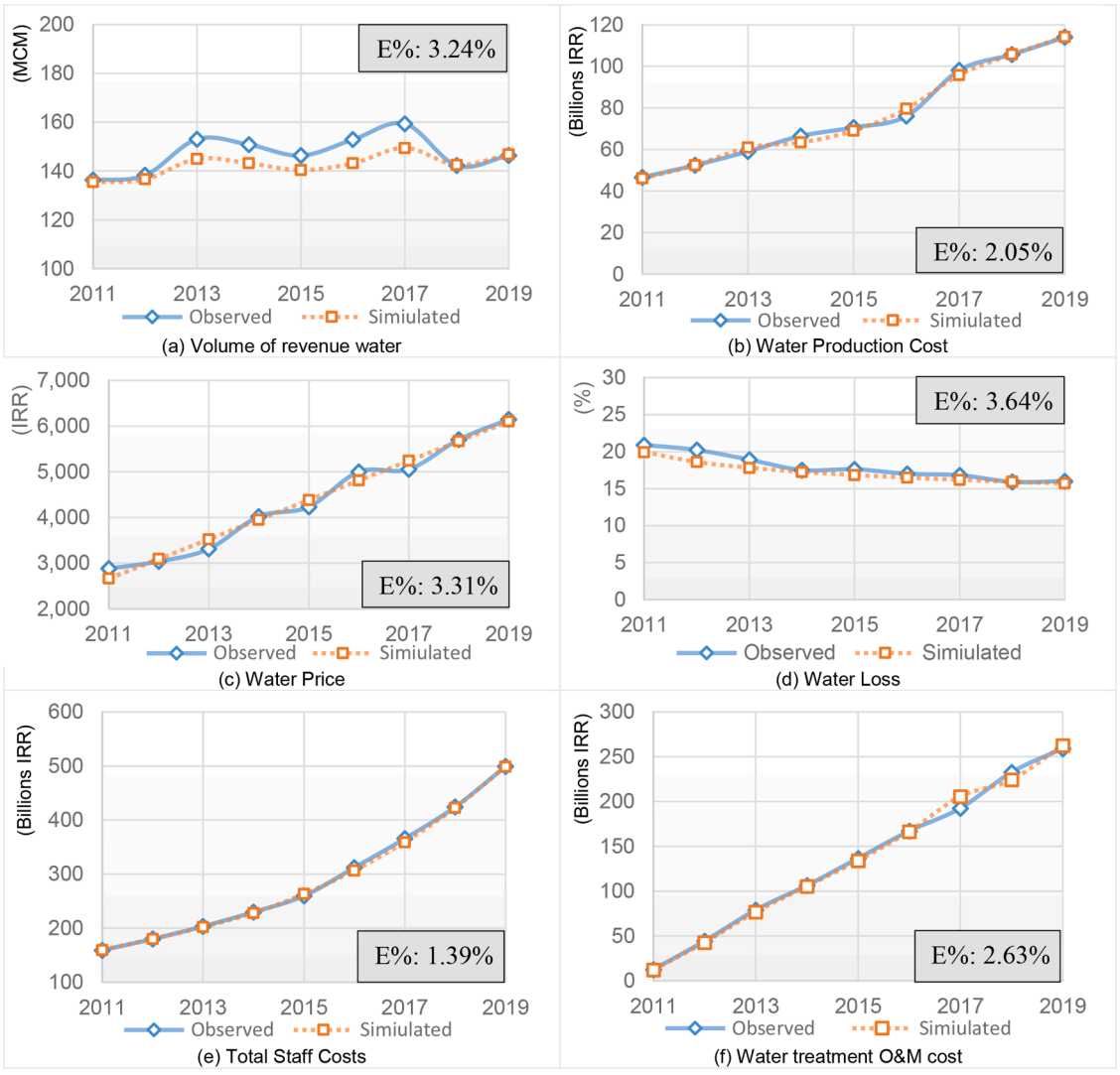


Fig 3. SDUWMM performance evaluation.

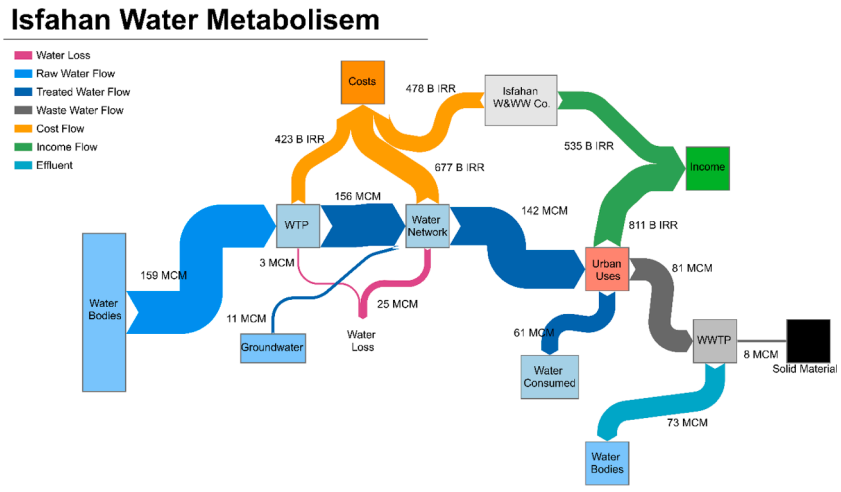


Fig 4. Isfahan Urban water metabolism diagram.

years, water demand has increased and the supply of water will not meet this demand. In addition, due to high inflation rate (average 18%), costs have increased at a lot of speed and the CCI ratio will be reduced to about 0.2. Therefore, despite these conditions that represent an inappropriate state of water resources and financial resources in the future, it is necessary to take new strategies to improve conditions.

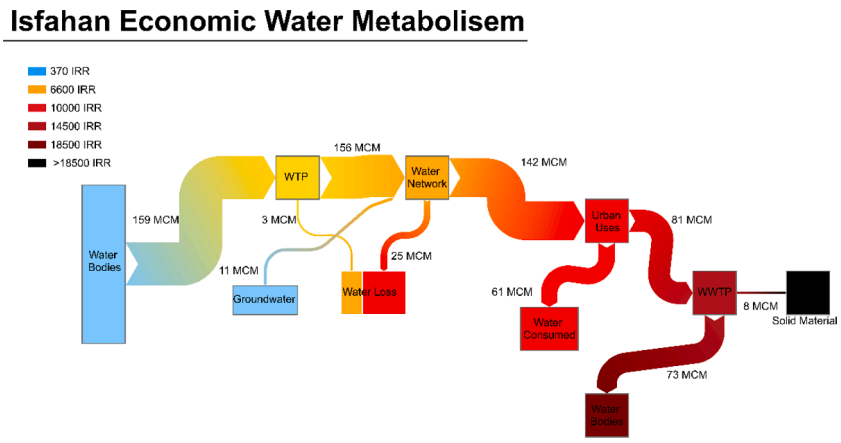


Fig 5. Isfahan economic water metabolism diagram.

4.2.1.2. Economic scenario. This scenario examines the effects of water prices on the revenues of IWWC and water consumption. In the EC1 scenario, the water price increased at an average rate of 15% annually. In the EC2 scenario, the water price rose at an annual average rate of 15% and the water price elasticity is -0.15%. Other conditions are similar to BAU. The results of these scenarios are shown in Figs. 6 and 7.

The revenue in EC1 and EC2 in 2041, respectively, was about 8.5% and 8%, and at total, 10.66% and 10.44% more than BAU, but the cost is almost equal. Therefore, the CCI for EC1 and EC2 is better than BAU. In general, due to the low water price (6150 IRR in 2019), increasing water price is a logical and early-return solution, which can increase economic resources and enhance cost coverage. However, due to the low price elasticity of water, it does not lower consumption. It also seems that water companies prefer higher water sales, consumption, and higher revenue consequently.

4.2.1.3. Social scenario. Raising social awareness in the consumption of water is possible by providing public education, familiarizing the public with water saving methods, and informing the general public of the critical state of water resources. For this purpose, in 2018, in Isfahan province, the following methods were carried out for notification and training: Broadcast TV Teasers, live TV talk shows, meeting subscribers of high consumptionin IWWC, dispatching promotional teams to crowded areas of the city and distributing brochures, printing warning messages on the bills for subscribers of high consumption, printing comparative and analytical diagrams of the consumption of subscribers on their bills, giving speeches in student dormitories and schools, and holding consultation meetings with artists and teachers. The result of these measures was to reduce 20% of water consumption. Other conditions are similar to BAU. The modeling results of this scenario for Isfahan city between 2021 and 2041 are presented in Figs. 6 and 7. As observed, with social awareness, water consumption decreased by a total of 15% between 2021 and 2041. This is almost equivalent to water consumption of 5 years in Isfahan. Importantly, these measures are costly and the total cost is 3.1% more than BAU. By reducing water consumption, the amount of IWWC income from water sales is reduced by 3.8%. Despite these changes, the cost coverage rate does not differ and is about 0.5% less than BAU. In general, the results of educational and awareness-raising measures can be continually observed and its effectiveness is continuous.

4.2.1.4. Environmental scenario. In the environmental scenario, climatic status is simulated in the drought mode and the effects of drought on water consumption are considered (En). According to evaporation statistics during the period 1991 to 2017, precipitation and temperature data of GCM (General Circulation Model), using artificial neural network, evaporation and precipitation were simulated for 2020s and 2030s. The results showed that in the 2020s and 2030s, evaporation will increase by 3.8% and 5.1%, and precipitation will decrease by 1.7% and 3.2% respectively (Mirniamm et al., 2017). Using the Neural Net Fitting method in MATLAB software, water consumption has been modeled in future drought periods, which indicates increased water consumption in these conditions by 4.5% and 7.6%, respectively, in the 2020s and 2030s, respectively. In Figs. 6 and 7, the results of En scenario modeling are shown.

In the modeled conditions, it is observed that under climate change, water consumption will increase more than 4% over the next 30 years. With more water production, the cost will also rise, but the cost coverage will be almost the same as BAU.

4.2.1.5. Technical scenario. In the base conditions, the average network rehabilitation rate is approximately 0.75% per year, which is a small amount. In this scenario, due to the service life of infrastructures, the rehabilitation rate of the water network has been considered by 1% (T1a) and 2% (T1b) annually. This means that in T1a and T1b scenarios, respectively within 100 and 50 years, the entire network will be renovated. In another scenario, the effect of water-conserving fixtures in the kitchen, toilet and bath will be reviewed. The use of an aerator in kitchen taps and toilet faucet reduce 60% of water consumption. The use of low-consumption showerhead also reduces the consumption of water by about 40%.

In this scenario, it is assumed that the following equipment is used: a reduced flow showerhead, a tap aerator for kitchen tap, a tap aerator for basin tap, and a tap aerator for toilet tap. In total, by installing these fixtures, kitchen and bathroom water consumption are curbed by up to 30% to 40%, which will be a reduction of 16% for total water consumption. The cost of this equipment is also 2,600,000 IRR in 2021 (T2). The results of this modeling are shown in Figs. 6 and 7.

Technical strategies are usually costly and time-consuming. In T1A and T1b scenarios, by higher network rehabilitation rate, water loss decreases and prevents cost, energy and purified water loss.

Table 4 shows a summary of the defined scenarios.

4.3. Scenarios results

The scenarios' results, which are shown in Figs. 6 and 7, the population of Isfahan city will growth over time in the coming years and cause water demand increase. Population growth, as a social parameter, affect the technical, economic and environmental conditions of Isfahan. Increasing water demand requires more water production, which put

Table 4
Definition of Scenarios.

No.	Scenario	water price increase	water price elasticity	rehabilitation rate	social awareness	water conservation equipment	climate change
1	BAU	7%	0	0.75%	×	×	×
2	EC1	15%	0	0.75%	×	×	×
3	EC2	15%	-0.15%	0.75%	×	×	×
4	S	7%	0	0.75%	✓	×	×
5	En	7%	0	0.75%	×	×	✓
6	T1a	7%	0	1%	×	×	×
7	T1b	7%	0	2%	×	×	×
8	T2	7%	0	0.75%	×	✓	×

more pressure on water and environmental resources. In the next step, technical infrastructure should expand the production, transfer and distribution of water, which requires more expenses and energy. This situation burden more costs on water companies and increased GHG emissions. However, as shown in Fig. 7, due to the limited water resources, water supply is partially possible and then the city of Isfahan will face water shortages. To solve the shortage issue, S, EC2 and T2 scenarios focus on reducing water consumption. In this regards, scenario S was addressed the shortage issue using social tools. Moreover, scenarios EC2 and T2 addressed the issue by considering economic and technical tools, respectively. The results show that the raising social awareness measures is most effective compared with EC2 and T2. However, reducing water consumption by rising water prices indicates the relationship between the economy and the society. Moreover, the reduction of water consumption by the use of water-conserving fixtures indicates the relationship between the society and the technical subsystem. Under the climate change condition in Isfahan, the amount of rainfall will decrease, air temperature will increase, and water consumption will growth. According to Fig. 6-e, it was observed that water demand is higher than BAU. It indicates the influence of social behavior from environmental conditions. In line with the increase in water demand and production, energy consumption has increased and GHG emissions raised. This reflects the impact of technical and social subsystems on the environment.

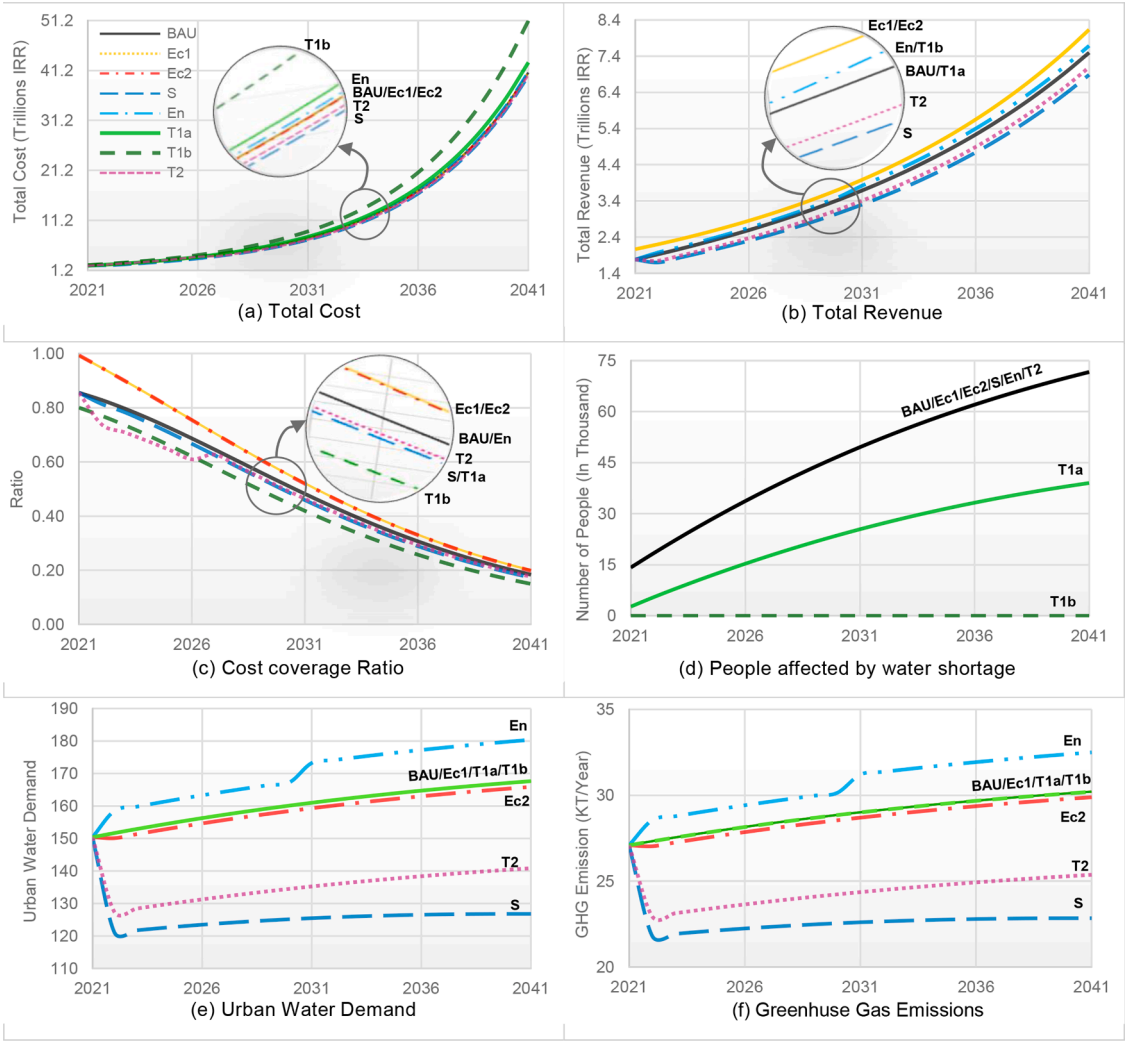


Fig 6. Results of Scenario Modeling.

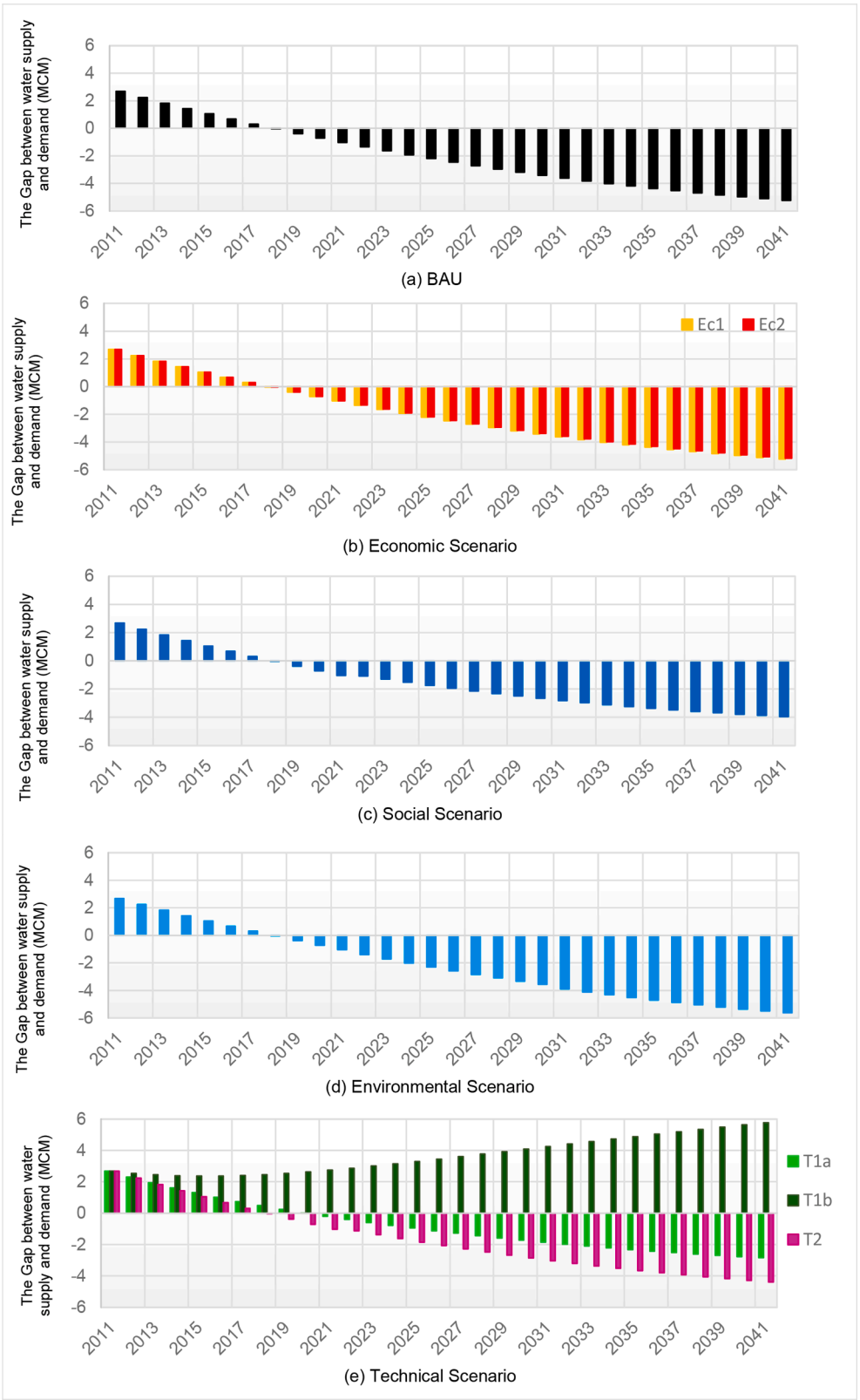


Fig 7. The Gap between water supply and demand simulated in model Scenarios.

4.4. Scenario evaluation

The Urban Water Sustainability Index (UWSI) was investigated in order to evaluate the performance of the scenarios in term of urban water sustainability. This index is influenced by CCI, WSRI and greenhouse gas emission index (GGI) in three pillars of sustainable development, economic, social and environmental, respectively. The values of parameters were normalized by Z-score normalization method and then Max-Min method was used to set data between 0 and 1. At the end for all scenarios, the UWSI (i.e., a geometric mean of three

Table 5
Results of scenario evaluation.

No.	Scenario	CCI	WSRI	GHGI	UWSI
1	BAU	0.5082	0.3912	0.3932	0.4276
2	EC1	0.8542	0.3912	0.3932	0.5084
3	EC2	0.8473	0.3912	0.4203	0.5184
4	S	0.3902	0.3912	0.9687	0.5288
5	En	0.5369	0.3913	0.2318	0.3652
6	T1a	0.4254	0.5494	0.3932	0.4513
7	T1b	0.1343	1	0.3932	0.3751
8	T2	0.3036	0.3912	0.8065	0.4575

indicators) has been calculated.

According to Table 5, scenarios that have the highest water sales (in terms of volume and price), EC1, EC2 and En, have the highest CCI. In fact, reducing water consumption reduces water companies' income and this is a significant point in water consumption management measures. In contrast, reducing water consumption reduces the volume of production and distribution of water, resulting in less energy consumed and lowering GHG emissions. Table 5 showed that the scenario S, which has the highest decrease in water consumption compared to BAU, has the lowest GGI and It is the most appropriate scenario environmentally. Improving water network condition and reducing non-revenue water, increase water supply reliability. As shown in the results, the T1A and T1B, which have the higher rehabilitation rate, have the higher WSRI. According to Table 5, the scenario S had the highest UWSI value. Due to the higher reliability and environmental-friendly of scenario S, it is the most sustainable and appropriate scenario; by excluding this scenario, scenario EC2, EC1 and T2 were the most reliable one in achieving urban water sustainability and covering the cost, respectively.

5. Conclusion

Using the modeling tool is a suitable solution for better recognition and decision-making. In this research, the SDUWMM model was developed from the combination of SD and UWM approaches to improve the status of urban water management in Isfahan. From a scientific and computational point of view, integration of the urban water metabolism and system dynamics creates a novel system that has been improved in terms of accuracy and simulation capabilities. In the field of engineering and practical application, the SDUWMM consider all dimensions of urban water sustainability include social, economic, technical, environmental, and governance and new relationships in the field of urban water metabolism have been developed that are more in line with the conditions of developing countries. Another innovation of this model is the definition of equations based on uses instead of resources that has been employed in previous developed models. In this model, the new economic flow is defined alongside the water flow, which makes financial management more transparent for water companies and helps decision makers to evaluate financial and water efficiency of their decisions, before executing them. The proposed UWM has two mainstreams, water and economics in the performance range of IWWC in the supply, treatment and distribution of water and wastewater collection. The SD model of Isfahan simulates the income and cost of IWWC and demand and water production in the future. Using these results, the gap between water production and required, and cost coverage rate were investigated. Scenarios were modeled for optimal use of funds and water, which had significant results. Actions on Supply Side, such as increasing rehabilitation rates, usually increase costs. However, demand management measures, such as social awareness and increasing water prices, usually have lower costs. Climate change in the Isfahan region will reduce precipitation and increase temperature, which leads to more water consumption. These conditions exacerbate the water crisis and push more pressure on the water and financial resources, which requires attention and preventive measures. Increasing water tariffs is one of the water demand management measures. With the implementation of economic scenarios due to low price elasticity of water, a slight change will occur in costs and consumption, but IWWC revenues will increase and the cost coverage will be improved. Installing Water Conserving fixtures reduces total water consumption by about 11%, and consequently reduces production, which reduces costs and reduces water scarcity. The results of the developed hybrid model and the analysis of scenarios show the ability to design and implement management strategies on the model, which can be used to achieve sustainable urban water management in Isfahan. Nevertheless, integrated urban water models require a large amount of accurate and reliable data. Since the data of different sources have been collected using various collections methods, the integration of such data causes incompatibility and difference (Paul et al., 2018). Therefore, collecting and preparing data before entering the model is an important step toward modelling. If the data with higher accuracy and detail is available, the model can be rebuilt seasonally, monthly or at the regional level rather than annually. This will be especially useful for examining the water shortage conditions, such as drought periods or different seasons of the year. The accuracy of the SDUWMM model increased with high quality normalized data in the water supply, distribution and consumption sectors, wastewater collection and treatment, and other sectors. This model also focuses on part of the urban water supply system that managed by urban water companies. To build a more comprehensive model, more details of the surface runoff and wastewater sector can be added to the model and developed precisely including their mathematical relationships in the next versions of the SDUWMM platform. This model can provide better dynamic insights for decision-makers to recognize urban water management issues and upgrade the status of financial and water resources in the results of decisions. In general, the use of UWM alongside the SD model improves the model's quantity and shows the effect of various actions and strategies with greater detail on UWM.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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